

Problem Statement

A small e-commerce online store maintains records of products manually.

Your task is to create a JavaScript program that stores product information and generates useful reports for the store manager. The program should use an array of objects to store product details and perform different operations using loops and conditional statements.

Product Information Store details of at least 15 products.

Each product should contain:

```
{  
    productId: 501,  
    title: "Leather Wallet",  
    price: 45.00,  
    category: "Fashion",  
    rating: 4.2,  
    isFeatured: true  
}
```

Task 1: Display All Products with Price Tags

Display complete information of every product available in the store catalog.

Expected Output Format

Product ID: 501

Title: Leather Wallet

Price: \$45.00

Category: Fashion

Rating: 4.2 Stars

Featured: Yes

Task 2: Calculate Total Valuation of Inventory

Calculate and display the combined total price of all unique product records available in the catalog.

Example

Total Catalog Value: \$850.50

Task 3: Identify Premium Products

Display only the products that cost more than \$100.00.

Example

Noise Cancelling Headphones - \$150.00

Smart Watch - \$220.00

Task 4: Apply Cyber Monday Discount

Loop through the products and display what the new price would be after applying a 10% discount.

Example

Leather Wallet - Original: \$45.00 | Promo Price: \$40.50

Gaming Mouse - Original: \$60.00 | Promo Price: \$54.00

Task 5: Count Featured Items by Category

Display how many "Featured" items exist in each product category.

Example

Electronics: 3

Fashion: 2

Home & Kitchen: 4

Task 6: Highest Rated Product

Find and display the product holding the highest customer review rating.

Example

Top Rated Product: Wireless Earbuds 4.9 Stars

Task 7: Customer Satisfaction Status

Display status according to rating scores.

Rating Score Status

4.5 or higher Highly Recommended

3.5 to 4.4 Satisfactory

Less than 3.5 Needs Review

Example

Wireless Earbuds - Highly Recommended

Leather Wallet - Satisfactory

Cheap Charging Cable - Needs Review

Task 8: Count Products by Satisfaction Tier

Display the number of products falling under each customer satisfaction status.

Example

Highly Recommended: 6

Satisfactory: 7

Needs Review: 2

Task 9: Display Featured Items Only Filter and display only the names of the products where isFeatured is true.

Task 10: Generate Store Performance Report

Display a final summary in the following format: STORE PERFORMANCE REPORT

Total Catalog Items: 15

Highly Rated Items: 6

Satisfactory Items: 7

Underperforming: 2

Max Product Rating: 4.9 Stars